CS3014 — Applied Human Computer Interaction
FAST National University of Computer and Emerging Sciences, H-11/4, Islamabad

D2: Synthesis Report, Problem Statement and Requirements

Item	Detail
Team	G10, MSM Technologies
Section	G
Stage	Define (Weeks 6–7)
Deliverable	D2, due Week 8
Project	Usability of the FAST-NU Insignia library portal: study room booking and book search
Submitted by	Muhammad Usman Al Haq (23I-0585), coordination lead for Define
Evidence base	Primary: D1B anonymised sessions P01 to P10, run 28 September to 1 October 2026. Supplementary: nine anonymous survey responses S01 to S09 (4 and 11 October 2026), used only to corroborate and listed in section 1.1b.

Name	Roll Number	Role (from the D0 charter)
Muhammad Umar	23I-0782	Team member; coordination lead for Prototype
Mirza Yahya Adeel	23I-0768	Team member; coordination lead for Empathize and Ideate
Muhammad Usman Al Haq	23I-0585	Team member; coordination lead for Define
Shahmeer Atif	23I-0711	Team member; coordination lead for Test

How to read this document

- Evidence IDs. P03-C2 means participant P03, Segment C (observed booking attempt), note 2. Segments follow D1A: A background, B retrospective booking, C observed booking attempt, D book search, E workarounds and closing. Every ID is listed in sections 1.1 and 1.1b. SV and S IDs are survey cards (section 1.1b).
- Other IDs. K = affinity cluster, T = theme, I = insight, N = need, U = user story, R = requirement, HMW = How-Might-We, POV = point-of-view statement. Numbers such as 2.3 inside the task analysis are task-analysis nodes.
- Quotations are exactly as typed in D1B or in the survey. [uncertain] is carried over from D1B where the note-taker and facilitator could not agree on what was said or done.
- Participants appear only as P01 to P10, and survey respondents only as S01 to S09. No names, roll numbers or identifying details appear.
- Nothing in this document describes a solution. Needs, opportunities and requirements are outcomes only.

Core task and scope. The core task is booking a library study room. D1A made it the primary research question (RQ1), and nine of the ten interview participants used or described a route that skips the portal for room booking (P07 is the exception). Book search (RQ2) is carried through the synthesis as theme T4 and kept as a secondary finding for later stages. It is not carried into the POV or the requirements.

Part 1: Synthesis

1.1 Evidence register

D1B session notes broken into one evidence unit per note (76 notes). Each row is one physical note on the affinity wall, written with its ID. The last column shows the cluster the note sits in on the final wall (clusters are defined in section 1.2). Cx is background, X is counter-evidence, O is outlier.

ID	Evidence note	Cluster
P01-A1	First semester. Uses the library three to four times a week.	Cx
P01-A2	Has logged into the portal once, at orientation.	K3
P01-B1	Last booked a room by asking at the desk.	K3
P01-B2	Did not know rooms could be booked online: "nobody told us."	K3
P01-C1	On her phone. First login failed, second worked.	K4
P01-C2	Said "where would rooms even be?" and took about three minutes to find the booking page.	K3
P01-D1	Searched for an introductory programming textbook by title. Said there were "too many versions."	K8
P01-D2	Used no filter. [uncertain] whether she noticed the mode selector.	K10
P01-E1	Without the portal she would "just walk to the shelf and look."	K9
P01-E2	Has never changed search mode.	K10
P02-A1	Second year. Books a room for team meetings about twice a month.	Cx
P02-B1	Sends a friend ahead to hold a room.	K1
P02-B2	Tried the portal once: "it said something about a slot and I didn't understand it." Abandoned it.	K5
P02-C1	Found the booking page quickly.	X
P02-C2	Picked the wrong date first and went back. Said "is this today or tomorrow?"	K5
P02-D1	Searched for a finance textbook on Google first: "Google gives me the author faster."	K9
P02-D2	Then used the portal search and found the book.	X
P02-E1	Asks in a WhatsApp group who has a free room.	K1
P02-E2	Did not know the search had modes.	K10
P03-A1	Second year. Uses the library mostly in exam weeks.	Cx
P03-B1	Booked online last semester: "I was never sure it went through."	K6
P03-B2	Went to the desk to check that booking.	K6
P03-C1	Reached the confirmation step without trouble.	X
P03-C2	Said "I'd normally screenshot this."	K6
P03-D1	Looked for an electronics reference. Clicked the second search option: "I think this is more exact." No results.	K8
P03-D2	Removed words from the query and found the book.	K8
P03-E1	Would ask a senior.	K9
P03-E2	Does not use filters.	K10
P04-A1	Third year. Books a room weekly for a study group and final year project meetings.	Cx
P04-B1	Rooms "disappear by ten in the morning," so he refreshes the page.	K2
P04-B2	Cannot see at a glance which rooms are free.	K2
P04-C1	Fast and practiced. Moved to the confirm button on the final screen and withdrew when the stop rule was repeated.	X
P04-D1	Searched by an author's name. Results included unrelated items. Said "this search is useless for authors."	K8
P04-E1	Asks a friend already inside the library to keep a room.	K1
P04-E2	Has asked the desk to hold one.	K1
P05-A1	Fourth year. Rarely uses the portal and prefers to study at home.	Cx
P05-B1	Last booked in second year: "I stopped bothering."	O
P05-C1	Could not find the booking page. Gave up after about four minutes. Logged as incomplete.	K3
P05-C2	Said "I'd just go to the desk."	K3

P05-D1	Searched for a journal article. Nothing relevant.	K8
P05-D2	Asked whether the portal "is only for physical books." [uncertain] whether a real belief or a guess.	K8
P05-E1	Uses Google Scholar and a shared drive kept by his department.	K9
P06-A1	Third year. Uses the library for case study preparation. Books rooms rarely.	Cx
P06-B1	Has never booked online. Walks in and "sees what's free."	K1
P06-C1	Unsure of credentials: "is it my roll number?" Worked on the second try.	K4
P06-C2	Found the booking page after scrolling the home page.	K3
P06-C3	Chose a three hour slot and asked "is there a limit?"	K5
P06-D1	Typed a full title with subtitle and got nothing. Shortened it and found the book: "I did not know that matters."	K8
P06-E1	Asks the librarian.	K9
P06-E2	Has used the portal only to check a due date.	Cx
P07-A1	Postgraduate, second semester, writing a thesis literature review. Uses the library daily and books rooms for thesis meetings.	Cx
P07-B1	"Booking is okay but the time slots are confusing." Segment B possibly primed (facilitator said "portal").	K5
P07-C1	Reached the confirmation step without help.	X
P07-C2	Noted that the earliest free slot is not shown.	K2
P07-D1	Needed a specific edition. Changed the search mode on purpose: "I know the advanced one but I don't trust it." Found the book on the second attempt.	K10
P07-E1	Uses Google Scholar and asks the thesis supervisor for sources.	K9
P08-A1	Part time MBA student. Visits the library on weekends.	Cx
P08-B1	Messages the desk to book, because "the portal is hard on mobile."	K7
P08-C1	On his phone. Said "this needs a lot of zooming."	K7
P08-C2	Mis-tapped the date field twice.	K7
P08-D1	Searched for a management book. Found it as the first result: "this was fine."	X
P08-E1	Does not use filters. Says he has never needed to.	K10
P09-A1	Second year. Had booked at the front desk earlier that day.	Cx
P09-B1	Tried online at 9 in the morning: "All rooms showed unavailable but I could see empty rooms."	K2
P09-B2	Went to the desk.	K1
P09-C1	Found the booking page.	X
P09-C2	At slot selection said "I don't know what this colour means."	K2
P09-D1	Looked for a lab manual. Asked the desk staff and did not use the search box for long.	K9
P09-E1	"The desk staff know which rooms are free."	K1
P10-A1	Third year. Books through the desk regularly for a project group of five.	O
P10-B1	Said the form "asked nothing about how many people." [uncertain] whether a capacity rule exists.	K5
P10-B2	Writes the room name on a sticky note for the group.	K6
P10-C1	Reached the confirmation step.	X
P10-C2	Said "I never know if it saved or not."	K6
P10-D1	Searched by course code and got nothing: "the portal doesn't know course codes."	K8
P10-E1	Shares booking details in the group chat.	K6

1.1b Supplementary survey evidence

Nine anonymous survey responses (S01 to S09; eight on 4 October and one on 11 October 2026) arrived after the D1B fieldwork. The survey was not part of the approved D1A protocol, so it is used only to corroborate and quantify the interview patterns. No theme, insight or need in this document rests on survey evidence alone. Respondents chose to answer, so the counts describe these nine students only. Free text is quoted exactly, including spelling. Denominators: 7 respondents needed a room (S01, S02, S03, S05, S06, S08, S09) and 6 needed a book and answered the search questions (S02, S03, S06, S07, S08, S09).

Respondent	Level	Programme area	Needed a room	Needed a book
S01	Undergraduate, Year 3-4	CS / SE / AI	Yes	No
S02	Undergraduate, Year 3-4	CS / SE / AI	Yes	Yes

S02	Undergraduate, Year 3-4	CS / SE / AI	Yes	Yes
S03	Undergraduate, Year 3-4	CS / SE / AI	Yes	Yes
S04	Undergraduate, Year 3-4	CS / SE / AI	No	No
S05	Undergraduate, Year 1-2	EE / CE	Yes	No
S06	Undergraduate, Year 3-4	CS / SE / AI	Yes	Yes
S07	Undergraduate, Year 3-4	CS / SE / AI	No	Yes
S08	Undergraduate, Year 3-4	CS / SE / AI	Yes	Yes
S09	Undergraduate, Year 1-2	Business	Yes	Yes

The survey broken into 32 notes, one idea each. SV cards are tick-box items counted across respondents; S0x-F cards are free-text answers from one respondent.

ID	Survey evidence note	Cluster
SV-1	5 of 7 respondents who needed a room ticked "Finding the booking page" (S01, S02, S05, S06, S09).	K3
SV-2	4 of 7 ticked "Choosing the date" (S05, S06, S08, S09).	K5
SV-3	6 of 7 ticked "Choosing the time or duration" (S02, S03, S05, S06, S08, S09).	K5
SV-4	5 of 7 ticked "Unsure whether the booking was saved" (S02, S05, S06, S08, S09).	K6
SV-5	4 of 7 ticked "Rooms showed unavailable but were not" (S03, S05, S06, S08).	K2
SV-6	1 of 7 ticked "Using it on mobile" (S06).	K7
SV-7	Of 6 who needed a book and answered the search question, 3 ticked "Too many results" (S03, S06, S08).	K8
SV-8	Of the same 6, 3 ticked "Irrelevant results" (S06, S07, S08).	K8
SV-9	Of the same 6, 1 ticked "No results" (S02).	K8
SV-10	Of the same 6, 3 ticked "Not sure which search mode to use" (S02, S06, S09).	K10
SV-11	8 of 9 answered No to "Did you know the portal has several search modes" (S01, S02, S03, S04, S05, S06, S08, S09). S07 answered yes and has used them.	K10
SV-12	Of 7 who needed a room, 5 booked via the portal (S01, S02, S05, S06, S08) and 2 asked at the front desk (S03, S09).	Cx
SV-13	Of 6 who needed a book, 2 start by asking library staff (S02, S09), 2 at the shelf (S03, S06) and 2 at portal search (S07, S08).	K9
SV-14	Asked what they have stopped trying on the portal, three wrote about books: S02 "Finding books", S09 "Finding a book", S06 "I stopped searching for books".	K9
SV-15	Of the same 6, 1 ticked "Found it quickly" (S07).	X
S02-F1	"so I ask the librarian and he uses the portal and then tells me where to look"	K9
S02-F2	"I still can't properly use the slider sometimes when booking" (writes that they are in the 7th semester).	K5
S02-F3	"I then go to the librarian for booking"	K1
S04-F1	"I don't ever go to the library"	Cx
S05-F1	"booked room but someone else had booked same room in same time slot"	K2
S06-F1	"the button to select the time duration wasent working and looked stale"	K5
S06-F2	Found out only "after lots of random clicking" that the date has to be selected before the time slot.	K5
S06-F3	"There was no need for selecting date, it was irrelevant."	K5
S06-F4	Says the interface "was not responsive".	K7
S06-F5	Search had "so many cluttered options and buttons" that they were "left confused what to do next".	K8
S06-F6	Search results "were not relevant to what i searched at all".	K8
S06-F7	Stopped searching for books because it is "more teadius to login and search, than to just ask the librarian to search it for me".	K9
S06-F8	"I also hesitate booking rooms as it requires too much effort"	O
S09-F1	"I couldn't understand the slider"	K5
S09-F2	Clicked the slider thinking the range was selected and it "kept getting unselecting bcz the annimation was weird".	K5
S09-F3	"there was no indication that my first click registered or not"	K5
S09-F4	A friend said to click once because "double clicks undo selection".	K5

How to read the survey responsibly: S02 ticked "Booked via the portal" but also writes that they go to the librarian for booking (S02-F3); both are recorded as given. S01 and S05 named a place they start looking for a book although they said they had not needed one, so those two answers are not used.

1.2 Affinity map: clusters

Clusters were formed bottom-up from the notes above and named only after the pattern was stable. Labels describe what is happening, not what to build. Note and participant counts in this table are D1B only; the next table adds the survey cards.

Cluster	Pattern label	Note IDs	Notes	Participants
K1	Students ask friends or the front desk, or walk in, to find out which rooms are free and to hold one.	P02-B1, P02-E1, P04-E1, P04-E2, P06-B1, P09-B2, P09-E1	7	4 (P02, P04, P06, P09)
K2	Students cannot read from the portal which rooms and slots are actually free.	P04-B1, P04-B2, P07-C2, P09-B1, P09-C2	5	3 (P04, P07, P09)
K3	Students did not know room booking exists, or cannot find where it starts.	P01-A2, P01-B1, P01-B2, P01-C2, P05-C1, P05-C2, P06-C2	7	3 (P01, P05, P06)
K4	Students need more than one attempt to sign in.	P01-C1, P06-C1	2	2 (P01, P06)
K5	Students are unsure what the date, slot and limit mean while they are choosing.	P02-B2, P02-C2, P06-C3, P07-B1, P10-B1	5	4 (P02, P06, P07, P10)
K6	Students are unsure a booking was recorded and keep their own proof.	P03-B1, P03-B2, P03-C2, P10-B2, P10-C2, P10-E1	6	2 (P03, P10)
K7	One student finds the booking form hard to operate on a phone.	P08-B1, P08-C1, P08-C2	3	1 (P08)
K8	Search queries return nothing or unrelated items, and students guess how to reword them.	P01-D1, P03-D1, P03-D2, P04-D1, P05-D1, P05-D2, P06-D1, P10-D1	8	6 (P01, P03, P04, P05, P06, P10)
K9	Students start looking for a book somewhere other than the portal.	P01-E1, P02-D1, P03-E1, P05-E1, P06-E1, P07-E1, P09-D1	7	7 (P01, P02, P03, P05, P06, P07, P09)
K10	Students do not use, or do not trust, the search modes and filters.	P01-D2, P01-E2, P02-E2, P03-E2, P07-D1, P08-E1	6	5 (P01, P02, P03, P07, P08)
X	Counter-evidence: found the booking page, reached confirmation, or got a result without difficulty.	P02-C1, P02-D2, P03-C1, P04-C1, P07-C1, P08-D1, P09-C1, P10-C1	8	7 (P02, P03, P04, P07, P08, P09, P10)
Cx	Background and context notes.	P01-A1, P02-A1, P03-A1, P04-A1, P05-A1, P06-A1, P06-E2, P07-A1, P08-A1, P09-A1	10	9 (P01, P02, P03, P04, P05, P06, P07, P08, P09)
O	Outliers: stopped using the portal for booking, or books by habit at the desk, with no reason stated or a different reason.	P05-B1, P10-A1	2	2 (P05, P10)

Notes per cluster on the full wall. The wall holds 108 notes: 76 from D1B and 32 survey cards.

Cluster	D1B notes	Survey cards	Total on the wall	Survey card IDs
K1	7	1	8	S02-F3
K2	5	2	7	SV-5, S05-F1
K3	7	1	8	SV-1
K4	2	0	2	none
K5	5	10	15	SV-2, SV-3, S02-F2, S06-F1, S06-F2, S06-F3, S09-F1, S09-F2, S09-F3, S09-F4
K6	6	1	7	SV-4
K7	3	2	5	SV-6, S06-F4
K8	8	5	13	SV-7, SV-8, SV-9, S06-F5, S06-F6
K9	7	4	11	SV-13, SV-14, S02-F1, S06-F7
K10	6	2	8	SV-10, SV-11
X	8	1	9	SV-15
Cx	10	2	12	SV-12, S04-F1
O	2	1	3	S06-F8
Total	76	32	108	

Clustering decisions (what was merged, split or left alone, and why).

- K1 and K2 were first seen as two groups, one about people and one about the screen. They are linked: the notes in K1 describe what students do about the problem in K2 (P09-E1 says the desk staff know which rooms are free). They are carried together as theme T1.
- K3 and K4 were merged into theme T2 because both are about reaching the booking page. K4 is only two notes and is not a theme on its own.
- K5 and K6 were kept as separate clusters (what the controls mean before choosing, and whether it was recorded after) and merged into theme T3 because in both students are unsure what the system did with their input.
- K7 stays a single-voice finding (P08 in the interviews, S06 in the survey). A single quote is not a theme, so it is not carried into a theme. It still informs requirement R5 and is marked as thin evidence there.
- K8, K9 and K10 are the book-search clusters. They are merged into theme T4 because all three describe students working around what search can find.

- P05-B1, P10-A1 and S06-F8 do not fit anywhere without inventing a reason, so they stay as outliers.
- X holds counter-evidence. It limits how far T2 can be generalised: six interview participants reached the booking page or confirmation step without difficulty (P02-C1, P03-C1, P04-C1, P07-C1, P09-C1, P10-C1).

Headline count. Nine of ten interview participants used or described a route other than the portal for getting a room: P01-B1, P02-B1, P03-B2, P04-E2, P05-C2, P06-B1, P08-B1, P09-B2, P10-A1. P07 is the one participant who described booking on the portal without a workaround.

1.3 From themes to insights to needs

The four strongest patterns. Each need is stated independently of any solution. The wording in the Insight and Need columns is the exact wording the POV statements in section 2.2 reuse.

Chain	Theme (what recurs)	Insight (why)	Need (what is required)	Clusters	Evidence IDs	Strength
T1 / I1 / N1	Students ask friends or the front desk, or walk in, to find out which rooms are free and to hold one.	I1: they cannot tell from the portal which rooms are actually free	N1: to know which rooms are free before they commit to a booking	K1, K2	P02-B1, P02-E1, P04-E1, P04-E2, P06-B1, P09-B2, P09-E1, P04-B1, P04-B2, P07-C2, P09-B1, P09-C2	12 notes, 5 participants (P02, P04, P06, P07, P09). Includes practiced users (P04, P07).
T2 / I2 / N2	Students who are new to the portal or book rarely did not know room booking exists, or cannot find where it starts, and need more than one attempt to sign in.	I2: they learn the portal can book rooms only if someone tells them, and they do not know where to look	N2: to know that room booking exists and where it starts	K3, K4	P01-A2, P01-B1, P01-B2, P01-C2, P05-C1, P05-C2, P06-C2, P01-C1, P06-C1	9 notes, 3 participants (P01, P05, P06). Holds for newer or occasional users only; see counter-evidence X.
T3 / I3 / N3	Students are unsure what the date, slot and limit mean while choosing, and unsure afterwards that a booking was recorded.	I3: they cannot be sure what the date, slot and limit mean, or whether a booking was recorded, so they ask aloud, check in person or keep their own proof	N3: to be sure what they have chosen and that the booking is held for them	K5, K6	P02-B2, P02-C2, P06-C3, P07-B1, P10-B1, P03-B1, P03-B2, P03-C2, P10-B2, P10-C2, P10-E1	11 notes, 5 participants (P02, P03, P06, P07, P10). If the possibly primed P07-B1 is removed, 10 notes and 4 participants remain.
T4 / I4 / N4	Students often start looking for a book outside the portal, and when they use portal search, queries that return nothing or unrelated items leave them guessing how to reword them.	I4: they do not know what the portal search can match, so they guess at rewording or look elsewhere	N4: to know what a search on the portal can find and how to ask for it	K8, K9, K10	P06-D1, P10-D1, P04-D1, P03-D1, P03-D2, P05-D2, P02-D1, P01-E1, P09-D1, P07-D1, P02-E2, P01-E2	21 notes, all 10 participants. Not carried into the POV (see scope note). Secondary.

Survey corroboration of the ladder. Supplementary only. The theme, insight and need wording above is unchanged.

Chain	Survey evidence	What it adds
T1	SV-5, S05-F1, S02-F3	4 of 7 respondents who needed a room ticked "Rooms showed unavailable but were not", the same mismatch as P09-B1. S05-F1 reports someone else booking the same room and slot.
T2	SV-1	5 of 7 ticked "Finding the booking page", against 3 of 10 interview participants (P01, P05, P06). Both samples are small.
T3	SV-2, SV-3, SV-4, S02-F2, S06-F1, S06-F2, S06-F3, S09-F1 to S09-F4	6 of 7 ticked time or duration, 4 of 7 the date and 5 of 7 "Unsure whether the booking was saved". Three respondents wrote about the slider or time control (S02, S06, S09), and S09-F3 says there was "no indication that my first click registered or not".
T4	SV-7 to SV-11, SV-13, SV-14, S02-F1, S06-F5 to S06-F7	8 of 9 did not know the portal has several search modes. 3 of 6 portal-search users were not sure which mode to use. 4 of 6 who needed a book start with staff or the shelf. Counter-evidence: S07 found a book quickly and uses the modes (SV-11, SV-15).

S06 wrote 8 of the 17 free-text notes, and no row above rests on S06 alone.

Findings that challenge the D0 assumptions. The evidence did not simply confirm D0.

- Assumption 1 (room booking) is only partly supported. Students stall at several points, not only the two D0 named: finding the page (P01, P05, P06), reading which rooms are free (P04, P07, P09), and the date or slot (P02, P06, P08, P10). In the interviews no participant named the "duration slider" that D0 blamed. They spoke about the "slot" (P02-B2, P07-B1), the date (P02-C2, P08-C2) and colour (P09-C2). The survey does name the control: three respondents wrote about the slider or time control (S02-F2, S06-F1, S09-F1) and 6 of 7 who needed a room ticked "Choosing the time or duration" (SV-3). So the control concern in D0 is supported by the survey and not by the interviews. Confirming was never observed because every interview participant stopped before confirming, by protocol.
- The bypass is real. Nine of ten interview participants used or described a non-portal route for getting a room (headline count in 1.2).
- Assumption 2 (search modes) is only partly supported. Four interview participants do not use filters or modes (P01-E2, P02-E2, P03-E2, P08-E1), but P08 found his book as the first result and called it "fine" (P08-D1), and P02 found hers on the portal (P02-D2). The stronger pattern is that students start elsewhere (K9, seven participants) and that queries return nothing (K8, six participants). The survey adds that 8 of 9 did not know the modes exist (SV-11).

1.4 Persona spectrum: students who book study rooms

The evidence shows real variation in how far students rely on the portal to get a room, so a spectrum represents it better than one composite persona. A single "average student" would hide the difference between P04, who books weekly and refreshes the page, and P01, who did not know rooms could be booked online.

Dimension: how far the student relies on the portal to secure a room. Portal bypassed (A) → portal tried, then checked elsewhere (B) → portal used by habit, still backed up by people (C).

Each point is a composite of several participants, not one person. Placement is a judgement from the notes: P02 sits at B because he attempted the portal and abandoned it (P02-B2) although he also sends a friend (P02-B1); P08 sits at A because he messages the desk instead of booking (P08-B1).

	A. Desk-and-friends booker	B. Hesitant portal booker	C. Practiced portal booker
Built from	P01, P05, P06, P08, P09, P10	P02, P03	P04, P07
Goal	Get a room for the group by whichever route works (P09-A1, P10-A1, P06-B1).	Book a room on the portal for team meetings or exam-week study (P02-A1, P03-A1, P03-B1).	Secure a room for regular study group, project or thesis meetings (P04-A1, P07-A1).
Behaviour	Asks at, messages or walks to the desk (P01-B1, P08-B1, P06-B1, P09-B2). Has used the portal for little else, for example checking a due date (P06-E2).	Tries the portal, then verifies elsewhere: sends a friend (P02-B1), asks a WhatsApp group (P02-E1), goes to the desk to check (P03-B2), would screenshot (P03-C2).	Books weekly or daily and is fast (P04-A1, P04-C1, P07-A1, P07-C1). Refreshes the page (P04-B1). Asks a friend inside or the desk to hold a room (P04-E1, P04-E2).
Frustration	Did not know booking exists (P01-B2). Could not find the page (P05-C1). All rooms showed unavailable while empty rooms were visible (P09-B1). Phone is hard (P08-B1).	A slot they did not understand (P02-B2). Never sure the booking went through (P03-B1).	Cannot see at a glance which rooms are free (P04-B2). Earliest free slot is not shown (P07-C2). Rooms gone "by ten in the morning" (P04-B1).
Context	New or occasional portal users (P01-A2, P05-A1, P06-A1). On a phone (P01-C1, P08-C1).	Booking about twice a month (P02-A1) or mostly in exam weeks (P03-A1).	Weekly (P04-A1) or daily library use (P07-A1). Competes for rooms early in the morning (P04-B1).
Needs	N2, N1	N3, N1	N1 (leading), N3
What would count as support	To find out that booking exists and where it starts, and to see which rooms are free without a person.	To be sure what they chose and that it is held for them.	To tell which rooms are free quickly, without refreshing or asking.

The need to tell which rooms are free (N1) shows up at all three points: A (P09-B1, P09-E1), B (P02-B1, P02-E1) and C (P04-B2, P07-C2).

1.5 Scenario

A composite of observed behaviour. No detail is added for realism.

Question	Answer (evidence)
Who	A student who books study rooms, at any point on the spectrum.
Task	Get a room for a group meeting that day (P04-A1, P07-A1, P10-A1).
Context	Morning, when rooms "disappear by ten in the morning" (P04-B1) and one student tried at 9 (P09-B1). Often on a phone (P01-C1, P08-C1).
What happens	See narrative below.

A student needs a room for a group meeting later that day. They open the portal on their phone and sign in, sometimes at the second attempt (P01-C1, P06-C1). If they have not booked before they look for where rooms are, scroll the home page, and may give up and go to the desk (P01-C2, P06-C2, P05-C1, P05-C2). On the booking page they cannot tell which rooms are actually free, so they refresh, message a friend or the group chat, or ask the desk (P04-B1, P02-E1, P04-E2, P09-B1, P09-B2). If they do pick a date and slot they check, correct or ask about what they chose (P02-C2, P06-C3), and afterwards they keep their own proof such as a screenshot or a sticky note, or go to the desk to check (P03-C2, P10-B2, P03-B2).

1.6 Journey map: booking a study room

Stages follow how the task actually unfolds in the notes. Emotions are limited to what was said or visibly observed. Opportunities are needs, not features.

Stage	Actions	Thoughts	Emotions	Pain points	Opportunities (needs)
1. Decide how to get a room	Asks at the desk (P01-B1), sends a friend ahead (P02-B1), walks in (P06-B1), messages the desk (P08-B1).	"nobody told us" (P01-B2). "I stopped bothering" (P05-B1). "the portal is hard on mobile" (P08-B1).	Resigned (self-report, P05-B1).	Students do not learn the portal can book rooms (P01-B2). Some stop trying (P05-B1).	Help students know, when they need a room, that the portal can book one (N2).
2. Reach the booking page	Signs in and retries (P01-C1, P06-C1). Scrolls the home page (P06-C2). Searches for the page for about three minutes (P01-C2) or about four minutes and gives up (P05-C1).	"where would rooms even be?" (P01-C2). "is it my roll number?" (P06-C1). "I'd just go to the desk." (P05-C2).	Uncertain (observed, P01-C2, P06-C1). Giving up (observed, P05-C1).	First login fails (P01-C1). Page found only after scrolling (P06-C2). Task abandoned (P05-C1). Survey: 5 of 7 ticked "Finding the booking page" (SV-1).	Help students know how to sign in and where room booking starts (N2).
3. Check which rooms are free	Refreshes the page (P04-B1). Asks a WhatsApp group (P02-E1) or a friend inside (P04-E1). Asks the desk to hold a room (P04-E2). Goes to the desk after rooms show unavailable (P09-B2). Reads slot colours (P09-C2).	"All rooms showed unavailable but I could see empty rooms." (P09-B1). "I don't know what this colour means." (P09-C2). "The desk staff know which rooms are free." (P09-E1).	Uncertain (observed, P09-C2). Practiced but working around it (observed, P04-C1, P04-B1).	Cannot see at a glance which rooms are free (P04-B2). Earliest free slot is not shown (P07-C2). Displayed status disagrees with the room (P09-B1; survey SV-5, 4 of 7).	Help students know which rooms are free before they commit to a booking (N1). **Leading opportunity.**
4. Enter date and slot	Picks a date, goes back after a wrong one (P02-C2). Mis-taps the date field twice (P08-C2). Zooms in on a phone (P08-C1). Chooses a three hour slot (P06-C3).	"is this today or tomorrow?" (P02-C2). "is there a limit?" (P06-C3). "it said something about a slot and I didn't understand it." (P02-B2). "this needs a lot of zooming." (P08-C1). Form "asked nothing about how many people." (P10-B1, [uncertain]). Survey: "no indication that my first click registered or not" (S09-F3).	Uncertain (observed questions, P02-C2, P06-C3).	Wrong date first (P02-C2). Two mis-taps (P08-C2). A slot not understood, so the portal was abandoned (P02-B2). Survey: 6 of 7 ticked time or duration (SV-3).	Help students understand what date, slot and limit they are choosing, including on a small screen (N3).
5. Confirm and keep a record	Reaches the confirmation step (P03-C1, P07-C1, P10-C1) and stops, by protocol. Would screenshot (P03-C2). Writes the room on a sticky note (P10-B2). Shares details in the group chat (P10-E1). Goes to the desk to check (P03-B2).	"I was never sure it went through." (P03-B1). "I never know if it saved or not." (P10-C2). "I'd normally screenshot this." (P03-C2).	Unsure (self-report, P03-B1, P10-C2).	Students are not sure a booking is recorded and build their own proof (P03-B1, P10-C2, P10-B2). Survey: 5 of 7 ticked "Unsure whether the booking was saved" (SV-4).	Help students be sure what they have booked and that it is held for them (N3).

Gaps marked, not filled. (1) Submitting the booking was never observed, because every interview participant stopped before confirming (D1A stop rule), so stage 5 rests on retrospective accounts (P03-B1, P10-C2), remarks at the confirmation screen and the survey. (2) Using the room after booking was not part of the protocol and is not mapped. (3) Time-to-locate is reported in D1B for P01 (about three minutes) and P05 (about four minutes) only. (4) No participant or respondent described a screen reader, magnification tool or other assistive technology.

1.7 Leading opportunity

Help students know which rooms are free before they commit to a booking (N1, stage 3). It is the strongest candidate for the problem statement because it is:

- clearly supported: 12 interview notes from 5 participants, mixing observed behaviour (P07-C2, P09-C2, P04-C1) with retrospective accounts (P09-B1, P04-B1, P02-B1), plus survey support (SV-5, S05-F1);
- present at all three spectrum points (A: P09, P06; B: P02; C: P04, P07);
- central to the core task, because it is the step where students most visibly leave the portal for people (P09-B2, P04-E1, P04-E2, P02-E1);
- phrased as a need. Many designs could answer it, and none is chosen here.

N2 and N3 stay as supporting needs and feed the supporting POV statements in section 2.2.

Part 2: Problem Framing

2.1 Hierarchical task analysis: booking a study room

Two levels below the goal. Sub-goals follow the journey map stages 2 to 5 and the actions follow its Actions column. Journey stage 1 (deciding how to get a room) happens before the portal task starts, so it appears as an exit in Plan 0 rather than as a sub-goal. D0 recorded the portal path as My Account, Booking, Book Room, but the D1B notes do not record click paths, so actions 1.3 and 1.4 are written only at the level the evidence supports.

Goal 0: A study room is held for the group on the chosen date and time.

Level	ID	Sub-goal or action	Evidence
Sub-goal	**1**	**Reach the booking page**	P01-C2, P05-C1, P06-C2
Action	1.1	Open the library portal	All observed attempts (P01-C1 to P10-C1)
Action	1.2	Sign in	P01-C1, P06-C1
Action	1.3	Look for room booking	P01-C2, P05-C1, P06-C2
Action	1.4	Open the booking page	P02-C1, P09-C1
Sub-goal	**2**	**Find out which rooms are free**	P04-B2, P07-C2, P09-B1
Action	2.1	Read the rooms and slots shown	P04-B2, P07-C2, P09-C2
Action	2.2	Refresh the page	P04-B1
Action	2.3	Ask a friend, the group chat or the desk	P02-E1, P04-E1, P04-E2, P09-B2
Sub-goal	**3**	**Enter the date and slot**	P02-C2, P06-C3, P08-C2
Action	3.1	Choose the date	P02-C2, P08-C2
Action	3.2	Choose the slot and its length	P06-C3, P07-B1
Action	3.3	Go back and correct an entry	P02-C2
Sub-goal	**4**	**Confirm and keep a record**	P03-B1, P10-C2
Action	4.1	Review the details entered	P03-C1, P07-C1, P10-C1
Action	4.2	Submit the booking	Not observed (stop rule)
Action	4.3	Keep or check a record of the booking	P03-B2, P03-C2, P10-B2, P10-E1

Plans (order, conditions and repeats)

Plan	How it runs
Plan 0	Do 1, then 2. If a suitable room is free, do 3, then 4. If no room appears free, repeat 2 or leave the portal and use the desk or a friend. If 1 is not completed, leave the portal and use the desk (P05-C2).
Plan 1	Do 1.1, then 1.2. If sign-in fails, repeat 1.2 (P01-C1, P06-C1). Do 1.3 until the booking page is found, then 1.4. If the page is not found after a few minutes, exit Plan 0 (P05 stopped after about four minutes).
Plan 2	Do 2.1. If status is unclear or every room shows unavailable, do 2.2 and 2.3, in either order and as often as needed, and return to 2.1 each time. Stop when a room is chosen or the portal is abandoned (P09-B2).
Plan 3	Do 3.1, then 3.2. If either entry is wrong, do 3.3 and repeat 3.1 and 3.2 (P02-C2).
Plan 4	Do 4.1, then 4.2, then 4.3. In 4.3 students use one or more of: screenshot, sticky note, group chat, asking the desk.

Observations on the task structure. (1) Three steps repeat: sign-in (1.2, P01 and P06), refreshing and re-reading availability (2.1 and 2.2, P04) and re-entering the date (3.1 and 3.3, P02). (2) Action 2.3 takes the student out of the portal altogether, and it is observed for four participants (P02, P04, P06, P09). (3) Two sub-goals have exits to the desk: sub-goal 1 (P05) and sub-goal 2 (P09). (4) Action 4.2 was never observed, and 4.3 comes from retrospective accounts and what participants said at the confirmation screen.

2.2 Point-of-view statements

The leading POV carries the leading opportunity (N1). Two supporting POVs carry the other needs from the journey map so that every requirement traces to a POV. All three reuse the exact wording of the user, need and insight from sections 1.3 and 1.4.

POV1 (leading). Students who book study rooms need to know which rooms are free before they commit to a booking because they cannot tell from the portal which rooms are actually free.

POV2 (supporting). Students who book study rooms need to be sure what they have chosen and that the booking is held for them because they cannot be sure what the date, slot and limit mean, or whether a booking was recorded, so they ask aloud, check in person or keep their own proof.

POV3 (supporting). Students who book study rooms need to know that room booking exists and where it starts because they learn the portal can book rooms only if someone tells them, and they do not know where to look.

POV trace table

POV	POV part	Exact words in the POV	Same words in the synthesis (where)
POV1	User	students who book study rooms	Persona spectrum title, section 1.4
POV1	Need	to know which rooms are free before they commit to a booking	Need N1, section 1.3 ladder table, row T1
POV1	Insight	they cannot tell from the portal which rooms are actually free	Insight I1, section 1.3 ladder table, row T1
POV2	User	students who book study rooms	Persona spectrum title, section 1.4
POV2	Need	to be sure what they have chosen and that the booking is held for them	Need N3, section 1.3 ladder table, row T3
POV2	Insight	they cannot be sure what the date, slot and limit mean, or whether a booking was recorded, so they ask aloud, check in person or keep their own proof	Insight I3, section 1.3 ladder table, row T3
POV3	User	students who book study rooms	Persona spectrum title, section 1.4
POV3	Need	to know that room booking exists and where it starts	Need N2, section 1.3 ladder table, row T2
POV3	Insight	they learn the portal can book rooms only if someone tells them, and they do not know where to look	Insight I2, section 1.3 ladder table, row T2

2.3 How-Might-We questions

HMW	Question	Why not too broad	Why not already a solution	Follows from the POV?
HMW1	How might we help students who book study rooms know which rooms are free before they commit to a booking?	Names one user group and one outcome at one step of one task.	Names no screen, feature or channel. A display, a staff-side change or a different booking process could all answer it.	Yes: same user and need as POV1.
HMW2	How might we make it possible for students who book study rooms to tell which rooms are actually free without asking someone who can see the rooms?	Same user and outcome as HMW1, with one condition taken from the observed workaround (K1).	The condition says what students should not have to do, not how. No design is implied.	Yes: same user as POV1 and the insight I1 ("they cannot tell from the portal which rooms are actually free").
HMW3	How might we help students who book study rooms be sure what they have chosen and that the booking is held for them?	One user group and one outcome at the choosing and confirming steps.	Names no confirmation message, receipt or reminder.	Yes: same user and need as POV2.
HMW4	How might we help students who book study rooms know that room booking exists and where it starts?	One user group and one outcome at the first step of the task.	Names no menu, link, banner or announcement.	Yes: same user and need as POV3.

Part 3: Requirements

3.1 User stories

Each story is tied to one spectrum point from section 1.4 and names the journey stage it belongs to. None describes a feature.

ID	Persona (spectrum point)	Story	Journey stage	Evidence
U1	A. Desk-and-friends booker	As a desk-and-friends booker, I want to start a room booking myself on my phone, so that I do not have to message or walk to the desk to get a room.	2. Reach the booking page	P01-B2, P01-C2, P05-C1, P05-C2, P06-C2, P08-B1, P08-C1, P09-B2
U2	C. Practiced portal booker	As a practiced portal booker, I want to tell which rooms are free when I open the booking page, so that I do not have to keep refreshing it or ask a friend or the desk to hold a room.	3. Check which rooms are free	P04-B1, P04-B2, P04-E1, P04-E2, P07-C2
U3	B. Hesitant portal booker	As a hesitant portal booker, I want to be sure of the date and slot I chose and that the booking is held for me, so that I do not have to screenshot it, go back to correct it or check at the desk.	4. Enter date and slot; 5. Confirm and keep a record	P02-B2, P02-C2, P03-B1, P03-B2, P03-C2

3.2 Requirement record

Requirements are outcomes the design must satisfy. Each has a type, a testable criterion in the form metric, threshold, target and test condition, and the exact evidence it traces to. Targets are the team's own choices, justified below the table.

ID	Type	Requirement (outcome)	Success criterion (metric, threshold, conditions)	Evidence it traces to	Story / persona
R1	Usability	Students who have not booked a room on the portal before can reach the room-booking page unassisted.	Percentage of first-time room bookers who reach the booking page unassisted within 2 minutes is at least 80%, in a moderated test of at least 10 students who have never booked a room on the portal, each starting from the portal address on their own phone. Pass: page reached, no help, within 2 minutes. Fail: help given, task abandoned, or over 2 minutes.	P01-B2, P01-C2, P05-C1, P05-C2, P06-C2 (T2, I2, N2)	U1 / A. Desk-and-friends booker
R2	Usability	Students can tell which rooms are free for a chosen day and time from the booking page alone.	Percentage of participants who correctly identify every free and every taken room and slot in each of three prepared booking views is at least 90%, with at least 10 participants mixed across spectrum points A, B and C, no help, no asking anyone and at most 30 seconds per view. Views: all rooms taken, one room free, several rooms free. Pass: no misidentified room in any view. Fail: any misidentification, help, or over 30 seconds on any view.	P04-B1, P04-B2, P07-C2, P09-B1, P09-C2, P09-E1, P04-E1, P04-E2 (T1, I1, N1)	U2 / C. Practiced portal booker
R3	UX	Students feel sure that a booking they have made is held for them.	Mean rating of at least 4.0 on the statement "I am sure this booking is held for me" (1 strongly disagree to 5 strongly agree), asked straight after a completed test booking, from at least 10 participants, with no individual rating below 3.	P03-B1, P03-B2, P03-C2, P10-B2, P10-C2, P10-E1 (T3, I3, N3)	U3 / B. Hesitant portal booker
R4	Usability	Students enter the intended date, slot and length correctly on the first attempt.	Percentage of participants who book a specified room, date and two hour slot correctly on their first attempt is at least 90%, with at least 10 participants on their own phones and a written task card. Pass: correct date, start time and length, no going back to correct, no help. Fail: any wrong entry, any correction, or help.	P02-B2, P02-C2, P06-C3, P08-C2, P07-B1, P10-B1 (T3, I3, N3)	U3 / B. Hesitant portal booker
R5	Accessibility	Students can read room and slot status and complete the booking on a small phone screen without relying on colour or zooming in.	Inspection walkthrough of stages 2 to 5 at a 320 CSS pixel width and 200% text size: 100% of 6 checkpoints pass in every specified walkthrough case. (1) No horizontal or two-way scrolling (WCAG 2.2 SC 1.4.10). (2) The booking can be completed with no pinch-zoom. (3) Date and slot controls have a target of at least 24 by 24 CSS pixels (SC 2.5.8). (4) Every room and slot status can be identified with colour removed (SC 1.4.1). (5) Text contrast is at least 4.5 to 1 (SC 1.4.3). (6) A screen reader (TalkBack or NVDA) announces each status and the booking result (SC 4.1.3). In addition, 0 of 5 phone participants pinch-zoom or mis-tap the date field.	P08-B1, P08-C1, P08-C2, P09-C2 (K7, K2). K7 rests on one interview participant and one survey respondent; no assistive-technology user was in either sample (see 3.4).	U1 / A. Desk-and-friends booker

Coverage check: usability (R1, R2, R4), UX (R3) and accessibility (R5). Every requirement has a criterion and evidence IDs.

Survey evidence for each requirement (supplementary; the IDs above are unchanged).

Requirement	Survey cards	What it adds
R1	SV-1	5 of 7 ticked "Finding the booking page".
R2	SV-5, S05-F1	4 of 7 saw rooms shown as unavailable that were not. One respondent reports a double booking.
R3	SV-4	5 of 7 ticked "Unsure whether the booking was saved"

R3	SV-4	5 of 7 ticked "Unsure whether the booking was saved".
R4	SV-2, SV-3, S02-F2, S06-F1, S06-F2, S09-F1 to S09-F4	4 of 7 ticked the date and 6 of 7 time or duration. Respondents describe a slider whose first click gave no sign it registered and whose second click undid the choice (S09-F3, S09-F4).
R5	SV-6, S06-F4, S06-F1	1 of 7 ticked mobile and the same respondent says the interface "was not responsive". This matches P08, so K7 now has two independent sources, each a single voice.

Why these targets.

- R1, 80% within 2 minutes. In D1B the two participants who struggled took about three minutes (P01-C2) and about four minutes before giving up (P05-C1). Two minutes is shorter than either. 80% leaves room for one miss in five testers, because first-time bookers are the group most likely to stall.
- R2, 90% and 30 seconds. P04 "cannot see at a glance" (P04-B2). Thirty seconds per view is the team's working definition of at a glance for a three-view set. The 90% matches the other usability targets, because a wrong read leads to a wasted trip or a lost room.
- R3, mean 4.0 and no rating below 3. On a five point scale 4 is agree. A rating of 3 or below is neutral or doubtful, which is the state P03 ("I was never sure it went through") and P10 ("I never know if it saved or not") described. D1 gave no numeric baseline, so this threshold is a target, not an improvement over a measured figure. The survey shows 5 of 7 unsure whether the booking was saved (SV-4).
- R4, 90% first attempt. In D1B at least 2 of 10 participants had a date problem (P02-C2, P08-C2), and in the survey 6 of 7 ticked time or duration (SV-3), so a clean first attempt is not the norm today. 90% asks for a clear improvement without demanding perfection.
- R5, 100% of checkpoints. The checkpoints are binary conformance checks, so the threshold is all of them. Passing them would show those six checkpoints are met and would not prove the whole system is accessible.

3.3 Traceability chain

Requirement	Story / persona	POV / HMW	Insight / need	Theme / cluster	Evidence IDs
R1	U1 / A. Desk-and-friends booker	POV3 / HMW4	I2 / N2	T2 / K3, K4	P01-B2, P01-C2, P05-C1, P05-C2, P06-C2, P01-C1, P06-C1; survey SV-1
R2	U2 / C. Practiced portal booker	POV1 / HMW1, HMW2	I1 / N1	T1 / K1, K2	P04-B1, P04-B2, P07-C2, P09-B1, P09-C2, P09-E1, P04-E1, P04-E2; survey SV-5, S05-F1
R3	U3 / B. Hesitant portal booker	POV2 / HMW3	I3 / N3	T3 / K6	P03-B1, P03-B2, P03-C2, P10-B2, P10-C2, P10-E1; survey SV-4
R4	U3 / B. Hesitant portal booker	POV2 / HMW3	I3 / N3	T3 / K5	P02-B2, P02-C2, P06-C3, P07-B1, P10-B1, P08-C2; survey SV-2, SV-3, S09-F1 to S09-F4
R5	U1 / A. Desk-and-friends booker	POV1 / HMW1 for colour (K2); POV3 / HMW4 for phone use (K7, a single-voice finding, not a theme)	I1 / N1	T1 / K2 (colour); K7 (phone)	P09-C2, P08-B1, P08-C1, P08-C2; survey SV-6, S06-F4

3.4 Limits and gaps in the evidence

- Sample. Ten interview students from one campus, half from computing programmes, recruited purposively (D1B 3.5). Booking frequency is self-reported. The patterns here are for testing, not for claiming about all students.
- No assistive-technology user. None of the ten interview participants or nine survey respondents described a screen reader, magnification tool or other assistive technology, and D1A did not screen for one. R5 is therefore grounded in the phone and colour observations (P08, P09, S06) and in WCAG checkpoints, not in evidence from disabled users. It should be re-tested with such users in the evaluation stage.
- Confirmation never submitted. All interview participants stopped before confirming, by D1A design. What the portal shows after confirming was not observed, and R3 rests on retrospective accounts (P03-B1, P10-C2), remarks at the confirmation screen (P03-C2) and the survey (SV-4).
- Possibly primed notes. P07-B1 was collected after the facilitator said "portal" (D1B Part 4, deviation 1). K5 keeps four D1B notes from three participants and T3 keeps four participants if it is removed.
- Uncertain notes. P01-D2, P05-D2 and P10-B1 are marked [uncertain] in D1B. None is the only support for a theme. Whether a room capacity rule exists is unknown, so no requirement depends on it.
- Time-to-locate is reported in D1B for P01 and P05 only, so the baseline for R1 rests on two participants.
- Single-voice finding. K7 (phone use) rests on P08 and S06 and is used only to shape R5, where it is flagged.
- Counter-evidence. Six interview participants reached the booking page or confirmation step without difficulty (cluster X), so T2 and R1 apply to newer and occasional users, not all students.
- Book search. T4 is synthesised but no POV or requirement is built on it. It is kept for the next stage.
- Survey. The nine survey responses sit outside the approved D1A protocol and are used only as supplementary corroboration. Respondents self-selected, n is 9, ticks are self-reported, and most denominators are 6 or 7. S06 contributes 8 of the 17 free-text notes. Inconsistent answers are recorded as given (see section 1.1b).

3.5 Submission checklist

Requirement	Where addressed
Affinity map with named, evidence-traced themes	Sections 1.1 to 1.3
Physical affinity wall and process recording (Stages 1 to 5)	Submitted separately through the course LMS. Sections 1.1 and 1.1b list the 108 notes for Stage 1. Participants appear only as P01 to P10 and S01 to S09.
One theme, insight and need per strong cluster, each traceable	Section 1.3
Persona spectrum, evidence-grounded, gaps marked	Section 1.4 and section 3.4
Scenario and six-column journey map, opportunities phrased as needs	Sections 1.5 and 1.6
Two-level HTA with plans	Section 2.1

POV word-for-word from the synthesis, with trace table	Section 2.2
HMW questions checked for breadth and for being a solution	Section 2.3
2 to 3 user stories, each tied to a persona	Section 3.1
3 to 5 measurable requirements across usability, UX and accessibility	Section 3.2 (R1 to R5)
Testable criterion and evidence for every requirement	Sections 3.2 and 3.3